5.7 Using Congruent Triangles

Name _____ Period ____ Date _____

Geometry 5.7 (30-minute plan) · Student sheet

I can…

- use congruent triangles to find a length I can't measure directly.
- use CPCTC in a proof.
- plan a proof that uses congruent triangles.

Vocabulary · fill in the missing word

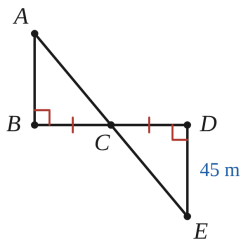
CPCTC — _____ parts of congruent triangles are congruent

Part 1 · Notes and examples (fill in each note, then work the example)

CPCTC: once two triangles are proven congruent, all of their _____ parts are congruent.

Example 1

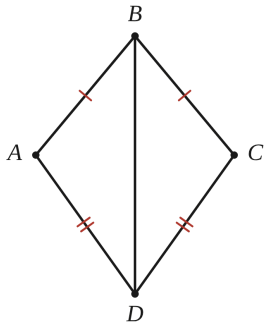
In the diagram, $\triangle ABC \cong \triangle EDC$ (ASA: right angles at B and D, $\overline{BC} \cong \overline{DC}$, vertical angles at C). DE = 45 m. How wide is the river (AB)?



Plan: to prove two parts congruent, find triangles that contain them, prove the triangles congruent _____, then use CPCTC.

Example 2

Given: $\overline{AB} \cong \overline{CB}$ and $\overline{AD} \cong \overline{CD}$. Prove: $\angle A \cong \angle C$.



Example 3

$\triangle PQR \cong \triangle STU$. PQ = 2x + 1 and ST = 15. $m\angle R = 40°$. Find x and $m\angle U$.

Part 2 · Partner practice

Partner practice

△JKL ≅ △MNO by SSS. m∠K = 72°. Find m∠N.